# Benjamin Liu

*Senior Software Engineer*

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## Summary

Senior Software Engineer with 10+ years of experience building scalable SaaS platforms across e-commerce, fintech, security, and healthcare, specializing in cloud-native architecture, distributed systems, and modern TypeScript/Node.js ecosystems. My key differentiator is designing secure, high-concurrency APIs and identity platforms (SSO, OAuth, JWT, RBAC) while driving performance through observability (logging, monitoring, tracing). I've led end-to-end delivery of systems supporting 5M+ daily operations with measurable reliability and growth impact.

## Work Experience

**PrimeiTek** — Northridge, CA

Senior Software Engineer — Feb 2020 – Jan 2026

- Led a team of 6 engineers delivering large-scale customer-facing platforms across eCommerce/Subscription, fintech payments, and HIPAA-aligned healthcare workflows, owning architecture, performance, and end-to-end delivery from discovery to production.
- Architected and delivered end-to-end full stack solutions using Next.js, React, TypeScript, Node.js, Express.js, MongoDB, PostgreSQL, and Redis, scaling platforms to 5M+ daily operations with high availability and sub-second API response targets.
- Led system design for microservices and platform foundations: API gateway patterns, caching strategies, connection pooling, and horizontal scaling across AWS/Azure, improving API latency by 32% through profiling, query optimization, and cache tuning.
- Owned large eCommerce and subscription initiatives (catalog → checkout → post-purchase), integrating Stripe and PayPal payment flows (tokenization, webhooks, retries, idempotency), plus ACH fund movement patterns aligned with PCI compliance and secure authorization standards.
- Delivered HIPAA-aligned healthcare capabilities by implementing secure access patterns, audit-friendly workflows, and privacy-conscious architecture across internal tools and patient-facing experiences, collaborating closely with compliance-minded stakeholders.
- Drove growth outcomes by improving acquisition funnel and onboarding journeys through analytics-driven iteration, performance monitoring, and experimentation, increasing onboarding conversion by 18% across key touchpoints.
- Built and maintained design systems and reusable component libraries (React/TypeScript), translating Figma into accessible, production-ready UI with measurable improvements to frontend performance and maintainability.
- Designed distributed backend workflows including scheduling, queuing, rate limiting, and serverless task processing to support high-volume JSON pipelines and workflow orchestration with strong retry/timeout/error-handling standards.
- Implemented enterprise-grade Identity & Access Management: JWT, OAuth, SSO integrations, RBAC, and scalable permissions modeling; delivered team management and role-based authorization patterns for multi-tenant environments.
- Improved reliability and operational readiness by implementing structured logging, monitoring, tracing, and automated alerting, reducing production incidents by 27% and shortening MTTR via better runbooks and dashboards.
- Established coding standards and clean code practices; ran structured code reviews, pair programming, and mentorship plans, elevating test quality with test automation, refactoring strategies, and consistent CI/CD enforcement.
- Partnered cross-functionally with Product, Design, and customer-facing teams to scope requirements, prototype high-risk components, and deliver a major platform release to completion with clear milestones and predictable delivery.

Key Technologies: Next.js, React, TypeScript, Node.js, Express.js, RESTful APIs, GraphQL, Microservices, Distributed Systems, AWS, Azure, Docker, Kubernetes, CI/CD, Redis, MongoDB, PostgreSQL, Serverless Architectures, Workflow Orchestration, Observability (Logging, Monitoring, Tracing), Identity & Access Management (SSO, OAuth, JWT, RBAC), Stripe, PayPal, ACH, PCI Compliance, Design Systems, Figma, Technical Leadership, Mentoring

**DroneDeploy** San Francisco, CA
Software Engineer Oct 2016 - Feb 2020

- Contributed to the full stack delivery of cloud-based reality capture, drone mapping, and geospatial analytics software used by customers in construction, agriculture, energy, and inspection workflows. The platform enabled automated flight planning, aerial data processing, and visual data insights by transforming raw drone imagery into 2D maps and 3D models for operational decision-making.
- Implemented scalable frontend features in React to support interactive viewing, annotation, and collaboration on geospatial maps and 3D site models, enhancing usability for inspection, progress tracking, and project documentation.
- Built backend services with Node.js and Express, implementing RESTful APIs for map processing, imagery uploads, and user-centric workflows across survey, construction, and monitoring use cases.
- Led database architecture enhancement and query optimization, improving performance for large-scale geospatial datasets and supporting collaborative storage and retrieval of high-resolution maps and models.
- Integrated cloud processing workflows to transform drone-captured aerial imagery into orthomosaics, digital elevation models, and analytics-ready geospatial outputs used for earthwork measurement, crop analysis, and site documentation.
- Implemented reverse proxy and traffic routing configurations using NGINX to improve system isolation and operational resilience across distributed cloud environments.
- Designed secure authentication and OAuth-based identity flows for enterprise customers, ensuring safe access to mission data and map assets.
- Partnered with DevOps to optimize AWS EC2, Auto Scaling Groups, and load balancing strategies for high availability and efficient processing workloads.
- Communicated architectural decisions and technical trade-offs with cross-functional partners in product, design, and data science to balance usability with performance constraints.

Key Technologies: React, Node.js, RESTful APIs, AWS (EC2, Auto Scaling, Load Balancing), NGINX, OAuth, Database Architecture & Query Optimization, Distributed Systems, Cloud Deployment, Geospatial Mapping & Analytics

**WLCM App Studio** San Francisco, CA
Full Stack Developer Oct 2013 - Oct 2016

- Built and shipped education platforms and analytics tools for early-stage SaaS products, supporting learning workflows, reporting, and performance dashboards from MVP to production.
- Implemented component-driven frontend development using modular, reusable UI patterns with JavaScript (ES5/ES6), HTML5, and CSS3, delivering scalable student, teacher, and admin experiences.
- Developed RESTful backend APIs with Node.js, Express.js, and PostgreSQL, implementing relational data models and business logic for course content, user activity, and reporting pipelines.
- Created interactive data visualization modules using D3.js for educational reporting and performance dashboards used by internal teams and customers.
- Applied Test-Driven Development (TDD) using Mocha/Chai to improve reliability of core modules and reduce regression risk during feature expansion.
- Managed CI/CD processes with Jenkins, supporting automated builds and AWS deployments for consistent releases across environments.
- Collaborated in Agile/Scrum teams, participating in sprint planning, backlog refinement, and iterative releases aligned with product goals.

Key Technologies: Angular, Node.js, Express.js, PostgreSQL, D3.js, RESTful APIs, TDD (Mocha/Chai), Jenkins CI/CD, AWS, Full Stack Development, Education Platforms

## Education

**University of California, Davis** 2009 - 2013
Bachelor of Science (BS), Computer Science Engineering

## Skills

**Frontend:**
React.js, Next.js, Vue.js, TypeScript, JavaScript, Redux, Context API, Zustand, HTML5, CSS3, Responsive Design, SPA Architecture, Progressive Web Apps (PWA), Frontend Development, Vite, Tailwind CSS, Sass/SCSS, shadcn/ui, Modern React Patterns, Design Systems, Component Libraries, Storybook, Frontend Performance Optimization, Clean Code Practices, Figma Collaboration, UI/UX Systems, Visual Design Implementation, MUI (Material UI), Digital Products Development, Interaction Design, Product Design Principles, User Experience (UX), UX/UI, UX Design, React Native, Mobile Development

**Backend:**
Node.js, Express.js, RESTful APIs, GraphQL, Backend Development, Workflow Orchestration, Distributed Task Processing, Scheduling & Queuing Systems, Rate Limiting, Serverless Architectures, Algorithms, Microservices Architecture, SOLID Principles, Data Structures, Design Patterns, Payment Systems Integration (Stripe, ACH), Fintech Platforms, SSO (Single Sign-On), Team Management Systems, Test Infrastructure, Linux/Unix Operating Systems, API Integration, OpenAPI/Swagger, Java, Python, Django, Go

**Forms & Validation:**
Formik, Zod, Schema Validation, Secure Input Handling

**Databases & Data:**
MongoDB, PostgreSQL, MySQL, Redis, Elasticsearch, Firebase/Firestore, Supabase, Database Architecture, Query Optimization, JSON Processing, Healthcare Data Standards (HL7, X12, EDI), Large-Scale Data Pipelines, Data Modeling, Indexing, Performance Optimization

**DevOps & Cloud:**
AWS, Azure, GCP, Azure AI Services, Docker, Kubernetes, Infrastructure as Code (Terraform), Continuous Integration, Continuous Deployment, CI/CD, GitLab CI/CD, DevOps Practices, Serverless Deployment, Structured Logging, Observability, Tracing, Enterprise AI Deployment, Git

**Testing & Quality:**
Test-Driven Development (TDD), Jest, React Testing Library, ESLint, Prettier, Automated Testing, Code Quality Enforcement

**Security:**
JWT, OAuth, RBAC, Session Management, API Security, PCI Compliance, Secure Fund Flows, Payment Authorization Standards, SSO, Identity & Access Management (IAM), Permissions Modeling, AI Security, Prompt Injection Prevention, LLM Security Considerations

**Architecture:**
Full Stack Architecture, Distributed Systems, System Design, High-Concurrency Systems, Scalability Engineering, Identity Systems, Authorization at Scale, Refactoring Practices

**Methodologies & Soft Skills:**
Agile, Scrum, Agile Software Development, Code Review, Technical Mentorship, Coaching, Ownership, Team Collaboration, Innovation, Strategic Thinking, Decision-Making, Adaptability, Self-Starter Mindset, Written & Verbal Communication, Feedback-Driven Development, Software Development Lifecycle (SDLC), Testing (Unit testing, Integration, Jest), Problem Solving, Advanced Analytical and Problem-Solving Techniques, Technical Research, AI Solution Architecture, Continuous Improvement, Ethics, Attention to Detail, Code Quality

**AI & Artificial Intelligence:**
AI-Assisted Development (GitHub Copilot, ChatGPT, Cursor), OpenAI API Integration, Claude API Integration, LLM Integration, Prompt Engineering, LangChain, LLM Orchestration Frameworks, AI Development & Platform Architecture, AI Agents, AI-Augmented Engineering Workflows

**Blockchain & Web3:**
Ethereum, Web3.js, Ethers.js, Smart Contract Integration, Cryptocurrency, DeFi, DEX Architecture